

Alternative Access to Regulatory Annotations

Session 4

T6: Making the most of cis-regulatory annotations for genome-scale analysis

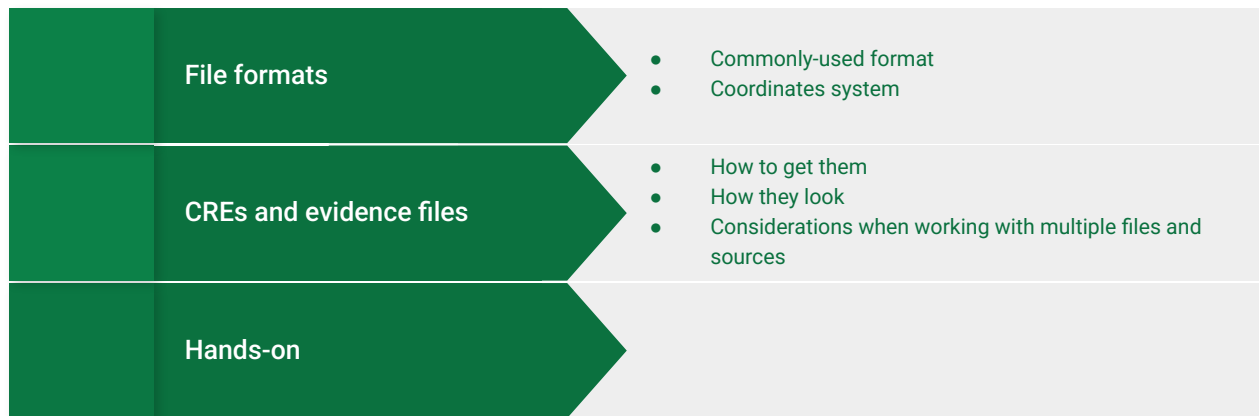
Gabriela A. Merino - Louisse Paola Mirabueno

Senior Bioinformatician,
Ensembl Regulation,
EMBL-EBI

Outreach Officer,
Ensembl Outreach,
EMBL-EBI

03/09/2026

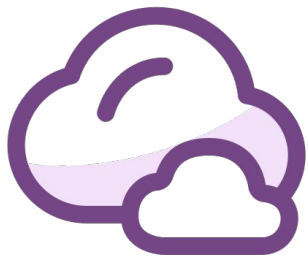
Session map





Have you used some of the regulatory annotation resources we have presented?





If you have previously used the regulatory annotation resources, how have you used them? E.g. single, locus analysis, file, etc.





For your future work, how do you think would be the best way to access them?



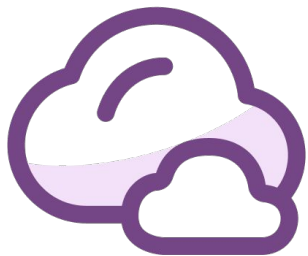
Commonly-used formats: BED and BigBed format

- ✓ Standard format for representing genomic intervals and annotations.
- ✓ It has three **required** fields and nine **additional** optional fields.
 - **chrom**: e.g. chr3, Y, NC_000001.11, scaffold10671.
 - **chromStart**: zero-based, included in the feature.
 - **chromEnd**: Not included in the display of the feature.
 - **name**
 - **score**
 - **strand**: Either "." (=no strand) or "+" or "-".
 - **thickStart**: The starting position at which the feature is drawn thickly (e.g., the start codon in gene displays).
 - **thickEnd**: The ending position at which the feature is drawn thickly.
 - **itemRgb**
 - **blockCount**, **blockSizes**, **blockStarts**: Blocks structure (i.e. exons)
- ✓ Uses 0-based, half-open coordinates:

E.g: the first 100 bases of chromosome 1 are defined as *chrom=1*, *chromStart=0*, *chromEnd=100*, and span the bases numbered 0-99.

Commonly-used formats: BED and BigBed format

- ✓ The number of fields per line must be consistent throughout any single set of data in an annotation track.
- ✓ The order of the optional fields is binding: lower-numbered fields must always be populated if higher-numbered fields are used.
- ✓ For very large BED datasets, UCSC recommends [BigBed](#), which provides indexing and faster remote access.



What type of track would you expect to be available as BED format?



Commonly-used formats: WIG and bigWig format

- ✓ Stores and displays continuous genomic signal data displayed as graphs in genome browsers.
- ✓ Designed for dense, **continuous** datasets with equally sized data elements.
- ✓ Supports two formats:
 - variableStep: irregularly spaced positions
 - fixedStep: regularly spaced positions (more compact)
- ✓ Structure:
 - Track definition line (track type=wiggle_0)
 - Declaration line (variableStep or fixedStep)
 - Data values (integer or floating-point, positive or negative)
- ✓ Important consideration: Unlike BED format, it uses 1-based, fully closed coordinates.
- ✓ UCSC recommends converting WIG files to bigWig for large datasets because it is indexed, faster to display, and more scalable.

E.g: the first 100 bases of chromosome 1 are defined as *chrom=1*, *chromStart=1*, *chromEnd=100*, and span the bases numbered 1-100.

Commonly-used formats: bedGraph format

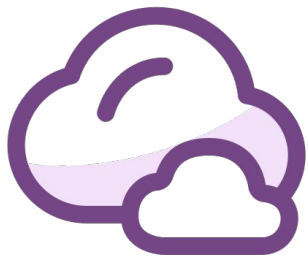
- ✓ Stores and displays continuous-valued genomic signals.
- ✓ Is similar to WIG, but preserves the original data values (no internal compression or binning).
- ✓ Suitable for both dense and sparse genomic signal data.
- ✓ Can store integer or floating-point values, including positive and negative numbers.
- ✓ Structure
 - Track definition line (track type=bedGraph)
 - Data lines with 4 columns: chrom, chromStart, chromEnd, dataValue
- ✓ It uses 0-based, half-open coordinates
- ✓ Large text files can become inefficient; UCSC recommends bigWig for large datasets.

E.g: the first 100 bases of chromosome 1 are defined as *chrom=1*, *chromStart=0*, *chromEnd=100*, and span the bases numbered 0-99.

Commonly-used formats: GFF3/GTF format

- ✓ Standard formats for representing genome annotations, including genes, transcripts, exons, CDS regions, UTRs, and other genomic features.
- ✓ Store both genomic coordinates and feature relationships (e.g., gene → transcript → exon).
- ✓ Support hierarchical annotation of complex gene structures.
- ✓ Human-readable, tab-delimited text formats containing 9 columns:
 - **seqid (chromosome/scaffold)**
 - **source**
 - **type (gene, exon, CDS, etc.)**
 - **start**
 - **end**
 - **score**
 - **strand**
 - **phase**
 - **attributes**
- ✓ Use 1-based, fully closed coordinates.

E.g: the first 100 bases of chromosome 1 are defined as *chrom=1*, *chromStart=1*, *chromEnd=100*, and span the bases numbered 1-100.



What type of track would you expect to be available as GFF format?



Commonly-used formats: GFF3/GTF format

- ✓ GTF (Gene Transfer Format)
 - Simpler format derived from GFF2.
 - Attributes typically include: gene_id, transcript_id, gene_name
- ✓ GFF3 (General Feature Format v3)
 - More expressive and standardized than GTF.
 - Uses explicit parent-child relationships:
 - ID=
 - Parent=
 - Supports complex annotations, including alternative transcripts, regulatory elements, and non-coding features.
 - Better suited for genome annotation projects and data exchange.
 - Official Sequence Ontology standard.

Commonly-used formats: Summary table

Format	Stores	Typical Use	Coordinates	Advantages	Recommended Large-Scale Version
BED	Genomic intervals/features	Genes, peaks, regulatory regions	0-based, half-open	Simple, compact, widely supported	bigBed
bedGraph	Quantitative signal values across intervals	Coverage, expression, scores	0-based, half-open	Preserves exact values; good for sparse or dense data	bigWig
WIG	Continuous signal data	Coverage tracks, probabilities, GC content	1-based, closed	Compact representation for dense data	bigWig
GTF / GFF3	Genome annotations with feature relationships	Genes, transcripts, exons, CDS	1-based, closed	Rich annotation structure; supports gene models	— (text-based standard)
bigBed	Indexed binary BED	Large annotation datasets	0-based, half-open	Fast access, efficient remote visualization	Current UCSC recommendation
bigWig	Indexed binary WIG/bedGraph	Large signal tracks	Depends on source format during conversion	Fast retrieval, scalable, browser-friendly	Current UCSC recommendation

Recommendations when working with multiple files

- ✓ Before analysis, write down:
 - ✓ Where each file came from
 - ✓ Which release or registry version it represents
 - ✓ Which filters were applied before download
 - ✓ Which coordinate system do the file uses
 - ✓ Which assembly has been used to produce it

source	<i>Ensembl / SCREEN / ENCODE / FAANG</i>
version	<i>release, registry version, retrieval date</i>
assembly	<i>GRCh38, GRCh37, mm10, ARS-UCD, etc.</i>
file	<i>URL, accession, checksum if available</i>
filters	<i>feature class, tissue, assay, status, cell type</i>
coordinate system	<i>BED or GFF convention; seqname style</i>

ENCODE cCREs annotation file



DownloadsAboutHelp

Enter a gene, cCRE, variant or locus

AnnotationsData MatricesDownload cCREs in Genomic Region

Human

cCREs by Class

cCREs by Cell and Tissue Type

cCRE-Gene Links

Functional Characterization

Mouse

Other

**Human (GRCh38/hg38)**
2,348,854 cCREs • 1,888 cell types

[Download Human cCREs](#)

Please cite our latest publication when using downloaded data: [Moore...Weng \(2026\) Nature](#)

cCREs by Class (.bed)

Promoter-like (PLS) (47,532)	2.6 MB	
All Candidate Enhancers (pELS & dELS) (1,718,669)	94.4 MB	
Proximal enhancer-like (pELS) (249,464)	13.7 MB	
Distal enhancer-like (dELS) (1,469,205)	80.7 MB	
Chromatin Accessible with CTCF (CA-CTCF) (126,034)	7.3 MB	
Chromatin Accessible with H3K4me3 (CA-H3K4me3) (79,246)	4.8 MB	
Chromatin Accessible with TF (CA-TF) (26,102)	1.5 MB	
Chromatin Accessible Only (CA) (245,985)	13.0 MB	
TF Only (TF) (105,286)	5.6 MB	
CTCF-Bound cCREs (948,642)	63.0 MB	
Silencer Sets (.tar.gz)	14.3 MB (unzipped)	
MAFF/MAFK Dynamic Enhancers (.tar.gz)	827 KB (unzipped)	

cCREs in Other Assemblies

GRCh37/hg19 (Lifted down from GRCh38/hg38)	128.8 MB	
T2T CHM13v2.0/hs1 (Lifted over from GRCh38/hg38)	123 MB	

ENCODE cCREs annotation file

```
head -n 6 GRCh38-cCREs.bed
chr1 10033 10250 EH38D4327497 EH38E2776516 pELS
chr1 10385 10713 EH38D4327498 EH38E2776517 pELS
chr1 16097 16381 EH38D6144701 EH38E3951272 CA-CTCF
chr1 17343 17642 EH38D6144702 EH38E3951273 CA-TF
chr1 29320 29517 EH38D6144703 EH38E3951274 CA
chr1 66350 66509 EH38D4327503 EH38E3951275 CA
```



Ensembl Regulatory Features annotation file

ENSEMBL © EMBL-EBI

Genome data & annotation Contact us

Regulation Human > Epigenomes Genome view Genes Regulatory features Data access Help & Docs About | Release: Latest

About the ENSEMBL project

GRCh38

Search...

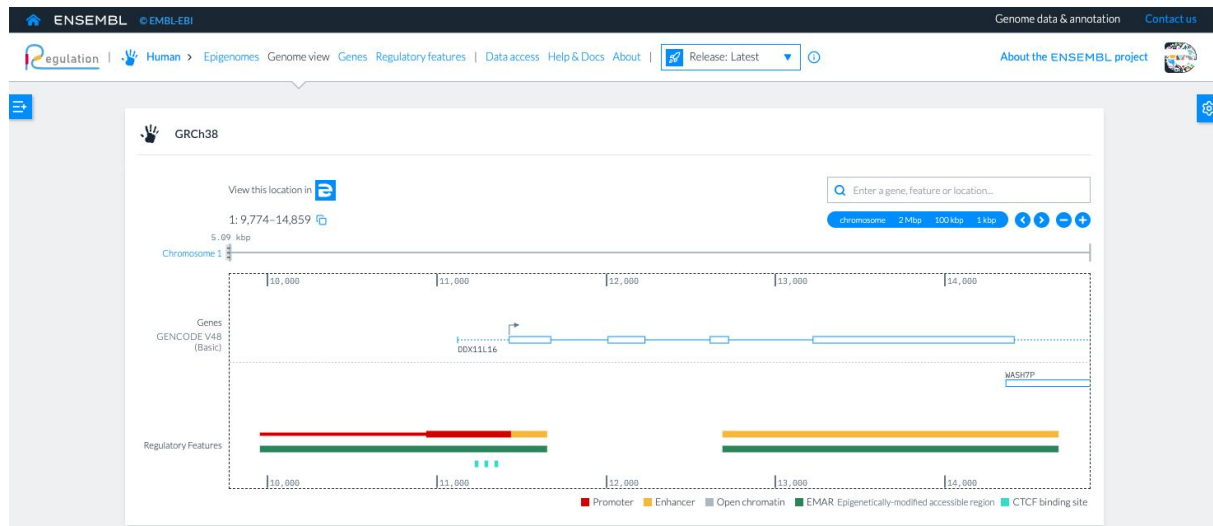
15 entries per page Showing 1 to 15 of 643528 features Download

ID	Type	Location	Genes
ENSR1_958	Promoter	1:10,936-11,436	DDX11L16
ENSR00001408561	EMAR	1:9,953-11,649	
ENSR1_53X2	CTCF binding site	1:11,222-11,243	
ENSR1_53X9	CTCF binding site	1:11,280-11,301	
ENSR1_53XJ	CTCF binding site	1:11,339-11,360	
ENSR1_88N	Enhancer	1:11,437-11,649	
ENSR00001408565	EMAR	1:12,686-14,671	
ENSR1_CZ	Enhancer	1:12,686-14,671	
ENSR00001408567	EMAR	1:15,101-15,853	
ENSR1_B48	Enhancer	1:15,101-15,853	
ENSR00001408568	EMAR	1:21,129-21,603	
ENSR1_986	Enhancer	1:21,129-21,603	
ENSR00001408569	EMAR	1:25,831-26,342	
ENSR1_B5Q	Enhancer	1:25,831-26,342	
ENSR1_9B7	Promoter	1:28,220-28,599	MIR1302-2HG

1

Ensembl Regulatory Features annotation file

```
head -n 8 Homo_sapiens.GRCh38.regulatory_features.v116.gff3
##gff-version 3
# Gene IDs in promoter attributes are from Ensembl release 114
1      Ensembl  promoter 10936      11436      .      .      .      ID=ENSR1_958;extended_start=9953;extended_end=11436;gene_id=ENSG00000290825;gene_name=DDX11L16;gene_biotype=lncRNA;color=#ff0000
1      Ensembl  CTCF_binding_site 11222      11243      .      +      .      ID=ENSR1_53X2;color=#40e0d0
1      Ensembl  CTCF_binding_site 11280      11301      .      +      .      ID=ENSR1_53X9;color=#40e0d0
1      Ensembl  CTCF_binding_site 11339      11360      .      +      .      ID=ENSR1_53XJ;color=#40e0d0
1      Ensembl  enhancer 11437      11649      .      .      .      ID=ENSR1_88N;color=#faca00
1      Ensembl  enhancer 12686      14671      .      .      .      ID=ENSR1_CZ;color=#faca00
```



Ensembl Regulatory Activity file

 **ENSEMBL** 

Genome data & annotation [Contact us](#)

 [Regulation](#) |  [Human](#) > [Epigenomes](#) [Genome view](#) [Genes](#) [Regulatory features](#) | [Data access](#) [Help & Docs](#) [About](#) |  Release: Latest 

[About the ENSEMBL project](#) 



Data access

Regulatory annotation

Ensembl's regulatory annotation is available as a TrackHub or for download.

TrackHub

The URL for the TrackHub is <https://regulation.ensembl.org/trackhub/hub.txt>. This will always reflect the latest release. If you wish to pin to a specific release, you can add the release to the URL e.g. <https://regulation.ensembl.org/2025-12/trackhub/hub.txt>.

Downloads

Regulatory features or filtered subsets can be downloaded from the [Regulatory features page](#). Regulatory activity can be downloaded [here](#).

Primary analysis files

Primary analysis track files (peaks and signal bigBeds) for selected epigenomes can be downloaded from the [Epigenomes page](#) by selecting bulk downloads.

Source data

The experimental datasets and associated metadata underlying the regulatory annotation are available through [Source data](#).

Ensembl Regulatory Activity file

```
head -n 2 Homo_sapiens.GRCh38.regulatory_activity.tsv
```

feature_id	feature_type	A549 (m, 58 y)	A673 (f, 15 y)	adrenal gland (f, 53 y)	...	vagina (f, 53 y)
ENSR1_958	promoter	ACTIVE	ACTIVE	...		ACTIVE
ENSR1_88N	enhancer	ACTIVE	ACTIVE	...		ACTIVE
ENSR1_CZ	enhancer	ACTIVE	ACTIVE	...		ACTIVE
ENSR1_B48	enhancer	ACTIVE	INACTIVE	ACTIVE	...	ACTIVE
ENSR1_986	enhancer	INACTIVE	INACTIVE	ACTIVE	...	ACTIVE

Regulatory activity 6 epigenomes

Epigenomes

A549 (m, 58 y)

A673 (f, 15 y)

adrenal gland
(f, 53 y)

adrenal gland
(m, 37 y)

adrenal gland
(m, 54 y)

vagina (f, 53 y)

Open chromatin: Peak Open chromatin: Signal H3K4me3 H3K27ac H3K4me1 H3K27me3

Ensembl Regulation Epigenomes Metadata

```
head -n 4 Homo_sapiens.GRCh38.metadata.2025-12.csv
epigenome,"biosample term","tissue group",sex,"life stage",age,"biosample material",assays
"A549 (m, 58 y)", "A549", "lung", "male", "adult", "58 y", "cell line", "ATAC-seq, CTCF, H3K4me3, DNase-seq"
"A673 (f, 15 y)", "A673", "muscle", "female", "child", "15 y", "cell line", "CTCF, H3K27ac, H3K27me3, H3K4me1, H3K4me3, DNase-seq"
"ACC112 (m, 70 y)", "ACC112", "epithelium", "male", "adult", "70 y", "cell line", "H3K27ac, H3K27me3, H3K4me1, H3K4me3"
```

ENSEMBL EMBL-EBI

Genome data & annotation [Contact us](#)

[Regulation](#) | [Human](#) > [Epigenomes](#) [Genome view](#) [Genes](#) [Regulatory features](#) | [Data access](#) [Help & Docs](#) [About](#) | [Release: Latest](#) ⓘ

[About the ENSEMBL project](#)

GRCh38

Showing 108 epigenomes

Epigenome	Biosample term	Tissue group	Sex	Life stage	Age	Biosample material
A549 (m, 58 y)	A549	lung	male	adult	58 y	cell line
A673 (f, 15 y)	A673	muscle	female	child	15 y	cell line
ACC112 (m, 70 y)	ACC112	epithelium	male	adult	70 y	cell line
adrenal gland (f, 53 y)	adrenal gland	endocrine gland	female	adult	53 y	primary tissue
adrenal gland (m, 37 y)	adrenal gland	endocrine gland	male	adult	37 y	primary tissue

EPIGENOMES
Metadata format: CSV
[Bulk download](#)

Ensembl Regulation Epigenome Peaks and Signals



GRCh38 epigenomes / A549 (m, 58 y)

base epigenome

ATAC-seq 3 Biosamples

Source	Biosample	Biomaterial type	Term	Life stage
ENCSR032RGS	ENCBS003HWJ	cell line	A549	adult
ENCSR032RGS	ENCBS049MEE	cell line	A549	adult
ENCSR032RGS	ENCBS024PJA	cell line	A549	adult

PRIMARY ANALYSIS FILES



PEAKS
format: bigBed



SIGNAL
format: bigWig



male 58 years

Questions?



Hands-on



Activity

Follow the example scripts that:

- ✓ Read Ensembl Regulatory Features and SCREEN cCREs subsets.
- ✓ Read user-defined regions and open chromatin peak files.
- ✓ Inspect loaded datasets.
- ✓ Overlap query regions with annotations.
- ✓ Filter by feature class.
- ✓ Identify specific features in the query regions.
- ✓ Define active features using open chromatin evidence peaks.



R packages you must

- **Rtracklayer**: Importing BED/GFF/GFF3 files
- **GRanges**: Storing, manipulating, and analyzing genomic intervals.
- **findOverlaps**: For variant or region intersection
- **dplyr**, **stringr**: Data manipulation with functions for filtering, selecting, summarizing, and arranging data
- **ggplot2**: highly-customizable data visualization



Python packages you must

- **PyRanges**: Storing, manipulating, and analyzing genomic intervals.
- **pandas, re**: Data manipulation and annotation parsing.
- **gffutils**: Efficient querying and management of GFF/GFF3 feature annotations.
- **matplotlib, seaborn**: Highly customizable data visualization.

